13 · Case 9, part 1: neurotransmitters and the synapse

The words of this track, as it says them. Bold brown is the examiner.

Exam case 9. Neurotransmitters and the brain.

Mr Paraone has been admitted to hospital after a fall, and reports problems with mobility and tremors. They're later diagnosed with Parkinson's disease: a brain disease where the cells that normally make dopamine, a neurotransmitter, start to die. One treatment is Levodopa, which travels through the blood and into the brain to boost the brain's dopamine levels, so the neurons can keep communicating properly via their synapses.

Two tracks. First, how a synapse works.

Two marks. What is a neurotransmitter? Where is dopamine normally made in the central nervous system?

A neurotransmitter is a chemical released at the presynaptic endings of a chemical synapse, which then acts on the receptor region of the postsynaptic neuron. Dopamine is made in the substantia nigra, in the midbrain, which is part of the brain stem.

Two marks. Explain the role of calcium ions in synaptic transmission.

When the action potential, the electrical nerve impulse, reaches the axon terminal of the presynaptic neuron, calcium channels open, and calcium enters the presynaptic neuron. The calcium makes the neurotransmitter vesicles open, releasing the neurotransmitter into the synaptic cleft, by exocytosis.

Six marks. Explain the process of synaptic transmission that allows neurotransmitters to be released. Use the diagram to help.

Six marks, six steps. The hook: arrive, calcium, release, cross, bind, respond, clear.

One. The action potential reaches the axon terminal of the presynaptic neuron, and calcium channels open, so calcium enters.

Two. The calcium makes the vesicles release the neurotransmitter into the synaptic cleft, by exocytosis.

Three. The neurotransmitter diffuses across the synaptic cleft.

Four. It binds to receptors on the postsynaptic neuron.

Five. The postsynaptic membrane's permeability changes, causing either an excitatory or an inhibitory effect.

Six. The neurotransmitter is removed, by re-uptake, or by enzymatic destruction.

Two marks. When neurotransmitters bind to a receptor, what happens?

The postsynaptic membrane's permeability changes, causing either an excitatory or an inhibitory effect. The postsynaptic neuron is either stimulated, or inhibited.

One mark. After the neurotransmitter has been released, what happens to that chemical signal?

It's removed. Either by re-uptake, back into the presynaptic neuron, or by enzymatic destruction. Take it back, or break it down.

Once more. The synapse.

A neurotransmitter is a chemical released at the presynaptic endings of a chemical synapse, which acts on the receptors of the postsynaptic neuron. Dopamine is made in the substantia nigra of the midbrain. The action potential reaches the axon terminal, calcium channels open, and calcium makes the vesicles release neurotransmitter into the cleft by exocytosis. It diffuses across, binds receptors on the postsynaptic neuron, and changes the membrane's permeability: excitatory, or inhibitory. Then it's cleared, by re-uptake or enzymatic destruction.